

Impact of Genotype on the Accuracy of an Image-Based Maize Leaf Angle Phenotyping Algorithm

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Introduction

Leaf angle is an important determinant of maize plant efficiency at intercepting light, influencing optimal planting density and thus maize yield. Higher yield due to optimal light interception can increase food availability, decreasing world hunger and/or releasing land for other uses. (Duvick 2005)

AngleNet, an automated phenotyping software, provides effective leaf angle measurements at scale (Xiang et al. 2023), though discrepancies exist between AngleNet measurements and historically collected manual measurements across maize genotypes. This project evaluates reliability of AngleNet by identifying trends and genotypic traits associated with discrepancies, providing information for improving AngleNet's accuracy and flexibility across genotypes for scalable phenotyping and support advancements in maize breeding.

Research Question

How do genotypic differences in maize affect the accuracy of the leaf angle measuring algorithm, AngleNet?

Null Hypotheses:

- There is no significant difference in the precision of AngleNet's leaf angle classification across the ten different evaluation orders of maize genotypes.
- AngleNet classifies leaf angles of EP1, Mo17, B73, W22, PH207, B104, F7, CML247, and PHW30 with the same precision.
- The presence of unusual structural traits, such as prominent shoot leaves or variations in leaf curvature, has no significant effect on the accuracy of AngleNet's leaf angle measurements.
- There is no significant linear relationship between a genotype's mean leaf angle and the number of incorrectly classified leaves.
- The distance of the leaf from the genotype mean ear height has no significant relationship with the accuracy of the leaf angle classification.

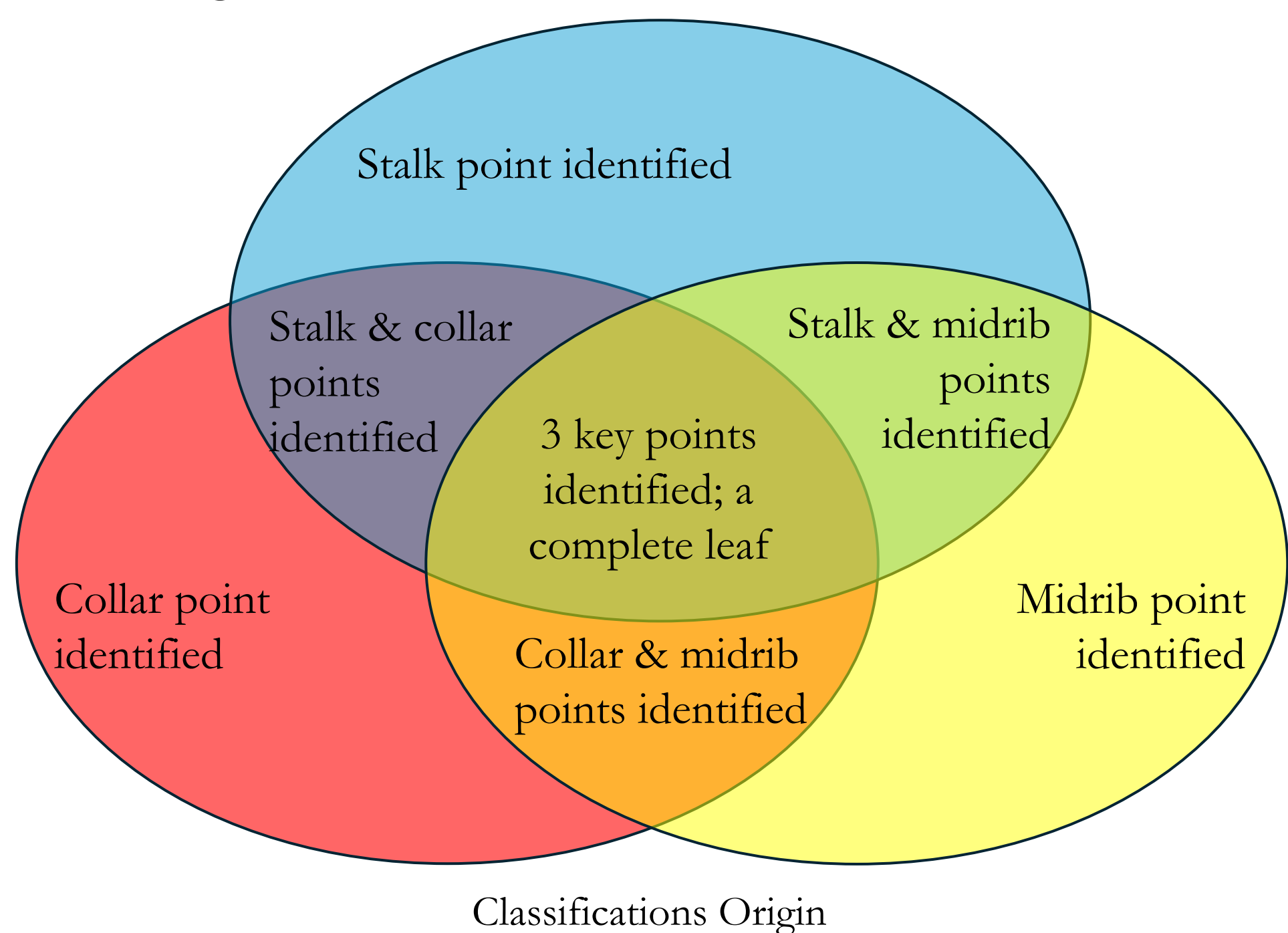
Materials/Methods

Materials

- Raw images of maize genotypes (EP1, Mo17, B73, W22, PH207, B104, F7, CML247, PHW30) captured by the AngleNet robot in summer 2023.
- AngleNet-derived measurements for each annotation: image ID, leaf height, and leaf angle from 3D images.
- Historical manual leaf angle measurements for genotypes.
- In-season plot-level average plant and ear height data for the Ames Diversity Panel.

Methodology Development

Classifications	Definition
(SCM)	All points correctly identified
(SC, M)	Midrib misidentified
(SM, C)	Collar misidentified
(CM, S)	Stem misidentified
(,SCM)	No point correctly identified
(S, CM)	Midrib & collar misidentified
(SC)	No midrib identified
(SM)	No collar identified
(CM)	No stem identified



Results

Hypothesis	Factor	Test	P-Value
1	Evaluation Order	Linear Regression	0.47998
2	Genotype	ANOVA	0.965
3	Genotypic Traits	T-Test on EP1	0.04254
4	Genotype Mean Leaf Angle	Linear Regression	0.0677
5	Genotype Mean Ear Height	Linear Regression	0.07194

The smaller the p-value, the more likely the hypothesis that the factor does not make a difference will be rejected.

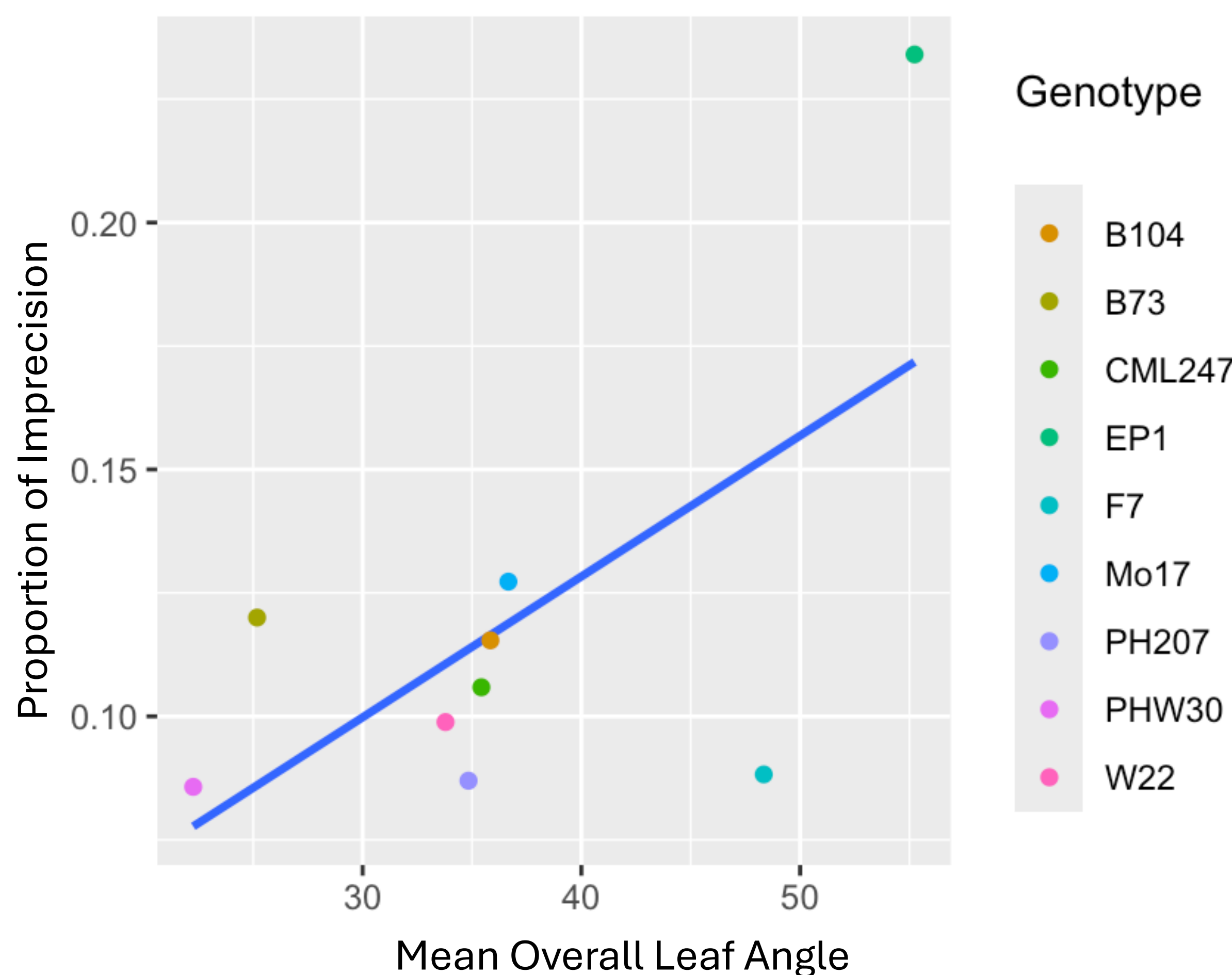


Figure 2: Scatter plot showing the relationship between a genotype's mean leaf angle and the imprecision of AngleNet's classifications. The blue regression line suggests a positive trend, indicating that as mean leaf angle increases, classification imprecision also tends to increase. (Hypothesis 4)

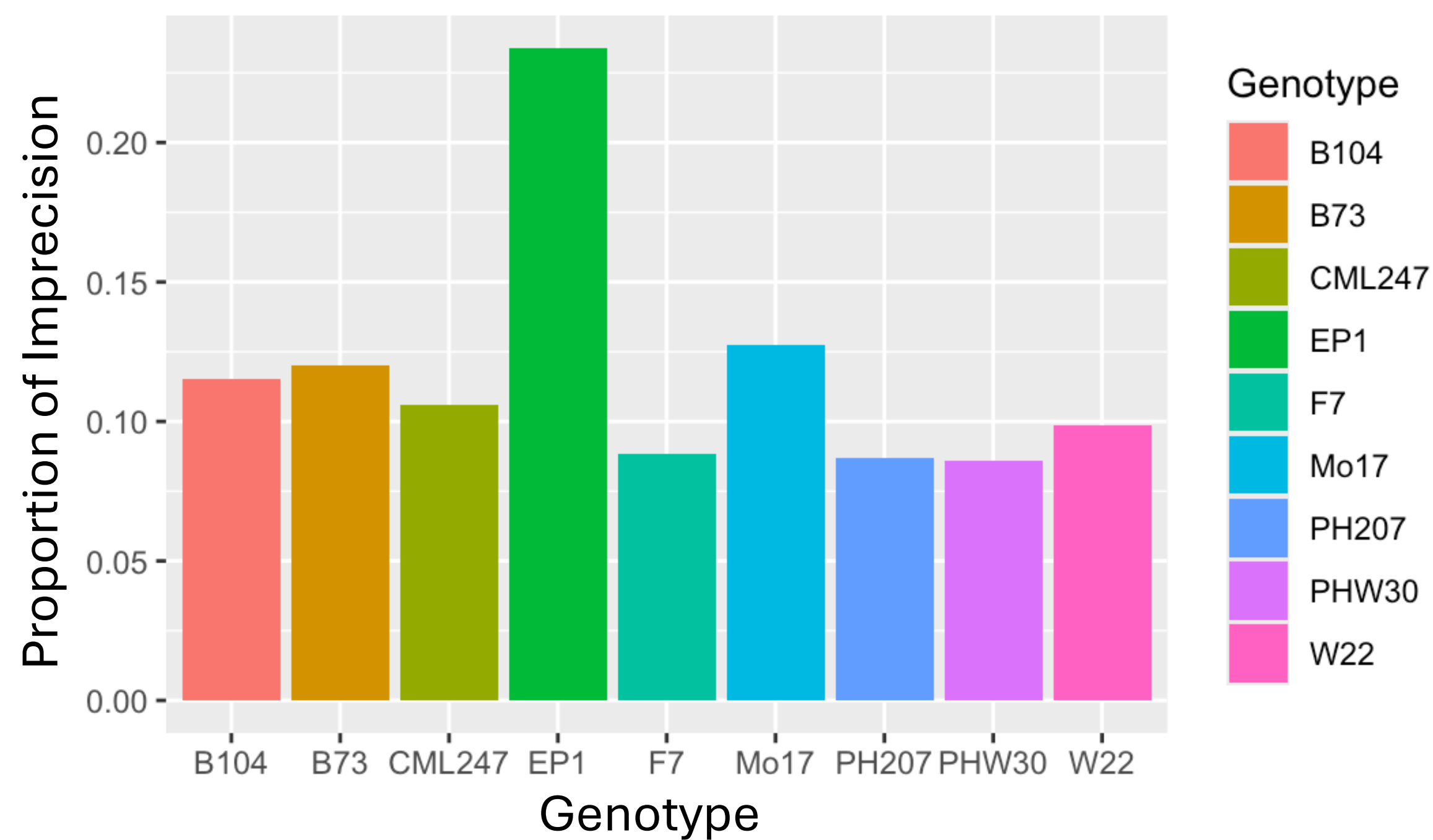


Figure 1: Column chart displaying the precision of AngleNet's leaf angle classifications for different maize genotypes. The results suggest that classification precision varies across genotypes, with EP1 showing the highest imprecision. This indicates that AngleNet does roughly classify all genotypes with the same level of precision except for EP1. (Hypothesis 2)

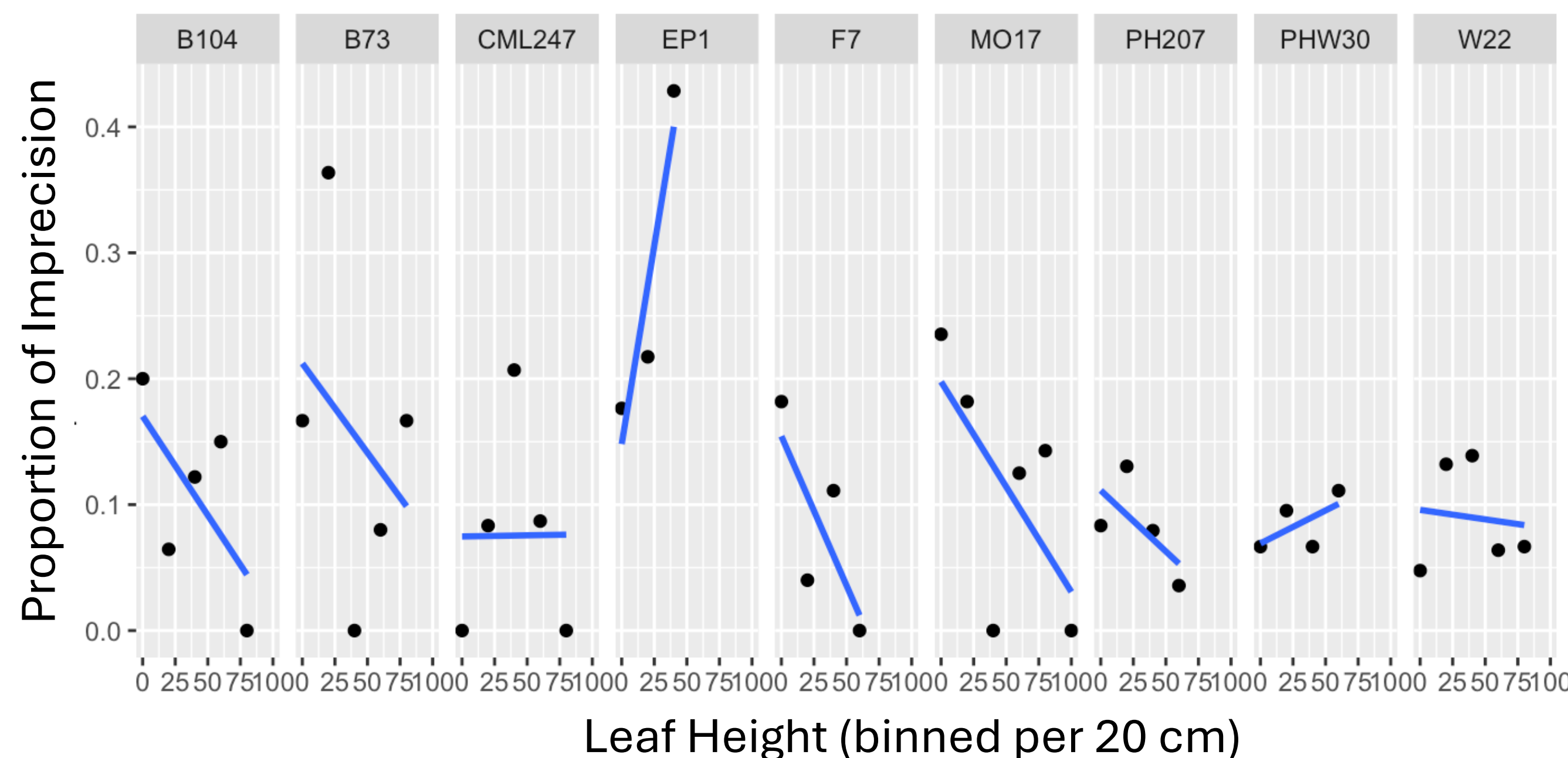


Figure 3: Precision of leaf classification across maize genotypes by height bin. Black dots represent precision values, and blue lines show linear trends. This tests whether leaf distance from mean ear height affects classification accuracy. (Hypothesis 5)

Conclusions

The results demonstrate that genotype influences the accuracy and consistency of AngleNet's leaf angle measurements, with certain genotypes exhibiting deviations due to their unique traits, such as mean leaf angle. Genotypes such as PH207, PHW30, and W22, which have more upright leaves (mean angle around 30°), exhibit lower imprecision values, whereas EP1, with less upright leaves and a higher mean angle, shows greater imprecision. This suggests that distance from the ear becomes a significant factor when leaves are flatter, but only in relation to certain genotypes.

Direct Implications: To improve AngleNet accuracy, further training on images of flatter leaves and those that clearly include the ear is necessary. Refining the model by incorporating more training data from genotypes with larger mean leaf angles, such as EP1 and PHW30, may also help enhance classification precision.

Greater Impacts: Leaf angle data remains a valuable trait for optimizing maize genotypes for efficient light capture. Improvements to AngleNet will not only enhance reliability but also expand opportunities with using human and land resources more effectively for global food production.

References

- Xiang, Lirong, et al. "Field-based robotic leaf angle detection and characterization of maize plants using stereo vision and deep convolutional neural networks." *Journal of Field Robotics* 40.5 (2023): 1034-1053.
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